

Patrick Meng
patrickmeng327@gmail.com

Education

Rutgers University, New Brunswick, September 2023 - Present

- PhD, Computer Science, Robotics

Columbia University, New York City, September 2021 - December 2022

- Masters, Computer Science, Machine Learning Track

Rutgers University, New Brunswick, September 2017 - May 2021

- Summa Cum Laude, Honors College, Highest Honors in Computer Science, Deans List
- Magison Award for Computer Science
- Mathematics and Computer Science Majors; Statistics Minor

Research Experience

Rutgers Pracsys Lab, September 2023-Present

Advisor: Kostas Bekris

- **Robot Camera Calibration**
 - Utilized foundation models to create accurate robot camera calibrations
- **Tensegrity Control**
 - Constructed tetherless tensegrity robot, a form of soft-rigid hybrid robot, from scratch
 - Published first real-world, closed-loop experiments on tensegrity robots

Computational Biomedicine Imaging and Modeling Lab, May 2024-Present

Advisor: Dimitris Metaxas

- **Diffusion based intra-oral data generation**
 - Collaborated with Colgate-Palmolive to generate synthetic images using diffusion methods

Columbia Roam Lab, January 2023-August 2023

Advisor: Matei Ciocarlei

- **Learned Decision Making with Tactile Sensors**
 - Contributed to the design, training, and testing of Deep Reinforcement Learning policies in Isaac Gym

Rutgers Pracsys Lab (Undergraduate), June 2017-August 2021

Advisor: Kostas Bekris

Teaching Assistant (TA) Experience

- Intro to AI (Fall 2023, Spring 2024), Intro to Comp. Rob. (Fall 2024), Data Sci. (Spring 2025)

Internships

Siemens, May 2022-August 2022

- **Autonomous train**

- Collaborated with LSL at Carnegie Mellon University to develop an autonomous train control system using Reinforcement Learning for the Federal Railroad Administration

Publications

J. Doerr*, **P. Meng***, S. Tangirala*, K. Bekris, “Integrating Explicit and Implicit Models for Robot Motion Generation,” *ISRR 2026*, Nov. 2026.. *Equal Contribution

N. Chen, **P. Meng**, C. Tang, A. DeGay, Z. Brei, R. Kramer-Bottiglio, K. Bekris, M. Aanjaneya,” Model Predictive Control of Tensegrity Robots via Contact-Aware Graph Neural Dynamics Model” *IROS 2026*, Sep. 2026

E. Granados, **P. Meng**, C. Tang, S. Sangani, W. Johnson, R. Kramer-Bottiglio, K. Bekris, “State and Trajectory Estimation of Tensegrity Robots via Factor Graphs and Chebyshev Polynomials,” *Robosoft 2026*, Apr. 2026

W. Johnson*, **P. Meng***, N. Chen, L. Cimatti, A. Vercoutere, M. Aanjaneya, R. Kramer-Bottiglio, K. Bekris, “[Demonstrating an Open-source, Reproducible System for Tensegrity Navigation among Obstacles](#)” *RA-L*, Mar. 2026. To be presented at IROS 2026, Sept. 2026. *Equal Contribution

J. Xu, Y. Jia, D. Yang, **P. Meng**, X. Zhu, Z. Guo, S. Song, and M. Ciocarlie, "[Tactile-based Object Retrieval From Granular Media](#)," *Autonomous Robots*, 49, Article 32. Aug. 2025
doi:10.1007/s10514-025-10212-9

K. Bekris, J. Doerr, **P. Meng**, S. Tangirala, “[The State of Robot Motion Generation](#),” *ISRR 2024*, Dec. 2024

P. Meng, W. Wang, D. Balkcom, and K. E. Bekris, “[Proof-Of-Concept Designs For the Assembly of Modular Dynamic Tensegrities into Easily Deployable Structures](#),” *Earth and Space 2021*, Sep. 2021. doi:10.26226/m.632b0aa4f30377bc3bafa226

H. Lim, A. Poleksic, Y. Yao, H. Tong, D. He, L. Zhuang, **P. Meng**, and L. Xie, "Large-Scale Off-Target Identification Using Fast and Accurate Dual Regularized One-Class Collaborative Filtering and Its Application to Drug Repurposing," *PLoS Comput Biol*, vol. 12, no. 10, pp. e1005135, Oct. 2016. doi: 10.1371/journal.pcbi.1005135.